

QMS Journal Club

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QMS Journal Club 7th meeting: What is “quantum”?

Quantum contextuality

What is considered as “physical”?

Before quantum theory is developed, the intuition that we have is that the physical properties of an object exist independent of observation.

Measurements merely act to reveal such physical properties.

According to quantum theory, an unobserved particle does not possess physical properties that exist independent of observation.

Such physical properties arise as a consequence of measurements performed upon the system.

In order to counter this argument, one proposed the non-contextual hidden-variable theory, where

- 1 All quantum observables may be simultaneously assigned definite values, which may deterministically depend on some “hidden” classical variable;
- 2 Value assignments pre-exist and are independent of the choice of any other observables;
- 3 Some functional constraints on the assignments of values for compatible observables are assumed.

Kochen and Specker proved that such hidden variable theory does not exist, i.e. it is impossible to assign definite values to an observable independent of other compatible observables (contexts).

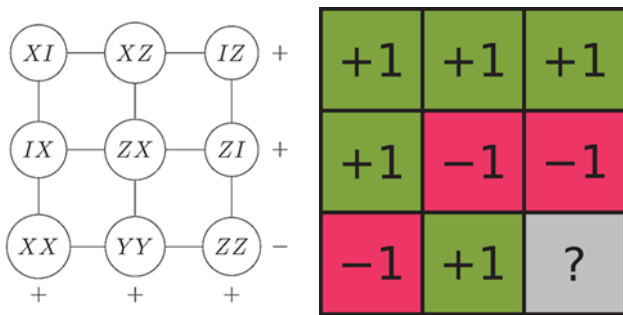


Figure 1: Mermin-Peres square.

Each line represents one context (mutually commuting operators) and each operator squares to identity, meaning that its eigenvalues are ± 1 .

The product of the operators on each line is $I_1 I_2$, except for the last vertical context.

There is no way to assign a definite value ± 1 to each operator to reproduce these product rules because each operator appears in exactly two contexts.

EPR paradox

In a complete physical theory, there is an element corresponding to each element of reality. A sufficient condition for the reality of a physical quantity is **the possibility of predicting it with certainty, without disturbing the system**.

In quantum mechanics, in the case of two physical quantities described by non-commuting operators, the knowledge of one precludes the knowledge of the other.

Then, either

- 1 The description of reality given by the wavefunction in quantum mechanics is not complete;
- 2 These two quantities cannot have simultaneous reality.

Bell's theorem

Bell's theorem assumes a local hidden variable model, whereby

- 1 “local” refers to the principle of locality, i.e. an object is influenced directly only by its immediate surroundings. In order for a cause at one point to have an effect at another point, something in the space between those points must mediate the action.
- 2 “Hidden variable” is a deterministic model to explain the probabilistic nature of quantum theory by introducing additional, possibly inaccessible, variables.

The correlations between the outcomes must obey a specific mathematical constraint, called Bell inequality.

Consider the following setup:

- 1 Alice and Bob stand in widely separated locations. A pair of particles is prepared. One particle is sent to Alice, the other to Bob.
- 2 Alice and Bob are free to choose two possible measurements, A_0 and A_1 , B_0 and B_1 , respectively. Each of these measurements produce ± 1 .
- 3 Suppose that each measurement reveals a property that the particle already possessed. For example, if Alice chooses to measure A_0 and gets $+1$, then the particle she received has $+1$ for a_0 .
- 4 Consider the combination for all possible properties a_0, a_1, b_0, b_1 :

$$a_0 b_0 + a_0 b_1 + a_1 b_0 - a_1 b_1 = (a_0 + a_1) b_0 + (a_0 - a_1) b_1. \quad (1)$$

- 5 Because a_0 and a_1 takes the values ± 1 , therefore it is either $a_0 = a_1$ or $a_0 = -a_1$. In the former case, $(a_0 - a_1) b_1 = 0$, while in the latter case, $(a_0 + a_1) b_0 = 0$. One of the terms on the RHS will vanish, and the remaining term will give ± 2 .
- 6 If the experiment repeats many times, the average will be given by

$$|\langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle| \leq 2. \quad (2)$$

Quantum entanglement violates this formulation of Bell inequality, called Clauser-Horne-Shimony-Holt (CHSH) inequality.

Consider the bell state,

$$|\psi\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}. \quad (3)$$

The first qubit is passed to Alice, while the second to Bob. Alice and Bob measure the following observables,

$$A_0 = \sigma_z, A_1 = \sigma_x; B_0 = -\frac{\sigma_x + \sigma_z}{\sqrt{2}}, B_1 = \frac{\sigma_x - \sigma_z}{\sqrt{2}}.$$

The expectation values are

$$\langle A_0 \otimes B_0 \rangle = \langle A_0 \otimes B_1 \rangle = \langle A_1 \otimes B_0 \rangle = \frac{1}{\sqrt{2}}, \langle A_1 \otimes B_1 \rangle = -\frac{1}{\sqrt{2}}.$$

Therefore, the CHSH inequality is violated,

$$\langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle = 2\sqrt{2}. \quad (4)$$

Quantum resource theory

In studying algorithms, three questions concern us:

- ① What resources are required to perform a given computational task?
- ② What computational tasks are possible?
- ③ What are the limitations (lower bounds) of the number of operations that must be performed by any algorithm?

Quantum resource theory studies quantum information processes under a restricted set of physical operations.

The permissible operations are “free”. Since they do not encompass all physical processes, only certain physically realizable states can be prepared. These accessible states are “free”.

The “prohibited” operations create “resource” states.

In quantum computing, entanglement is considered as a resource. The “free” operations are local quantum operation and classical communication (LOCC). Entanglement cannot be generated by LOCC.